Structural basis for red-shifted emission of a GFP-like protein from the marine copepod *Chiridius poppei*

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The fluorescence excitation and emission maxima of a GFP-like protein from the marine copepod *Chiridius poppei* (CpYGFP) show a significant red shift (λ_{ex} = 509 nm, λ_{em} = 517 nm) compared with those of GFP from *Aequorea victoria* (avGFP) and other GFP-like proteins from marine copepods. We performed crystallographic and biochemical studies to understand why this shift occurs in CpYGFP. The structure of CpYGFP showed that the imidazole side chain of His52 is involved in stacking on the phenol moiety of the chromophore. We investigated the potential role of His52 in causing the red-shifted spectral properties by performing mutational analyses of H52T, H52D and H52F. The emission wavelengths of H52T and H52D were blue-shifted and that of H52F was red-shifted relative to the wild type. Comparison of its structure of another copepod GFP (ppluGFP2) having an emission maximum at 502 nm showed that the imidazole ring of His54 (corresponding to His52 in CpYGFP) is flipped out of the stacking position with the chromophore. These findings suggest that π–π stacking interaction between His52 and the phenol moiety of the chromophore is the likely cause of the red-shift in light emission.

Introduction

The green fluorescent protein from the hydrozoan jellyfish, *Aequorea victoria*, (avGFP) has served as a versatile and useful marker for studying protein localization, protein trafficking and gene expression in live cells (Chalfie *et al*. 1994; Ogawa *et al*. 1995). The particular advantage of GFP is that a chromophore in its primary structure emits a greenish fluorescence upon irradiation with long UV light. The chromophore is post-translationally formed by autocatalytic modification of the tripeptide, X-Tyr-Gly, which is located in the primary structure of a precursor protein, without the participation of an external cofactor (Nishiuchi *et al*. 1998). Besides avGFP, numerous GFP-like proteins from hydrozoans and anthozoans have been isolated and characterized (Shagin *et al*. 2004), and they emit various colors of light (Verkhusha & Lukyanov 2004). Blue and cyan fluorescent variants are exceptional in that their chromophores are formed from X-His-Gly or X-Trp-Gly (Heim *et al*. 1994). Many mutation experiments with GFP-like proteins have yielded proteins with brighter fluorescence, various colors of light (Heim *et al*. 1994, 1995; Wachter *et al*. 1998), different pK_a values, dual emission (Hanson *et al*. 2002) and reduced oligomerization (Campbell *et al*. 2002). Several crystallographic studies of GFP proteins have revealed that chromophores are situated within the core of an 11-stranded β-barrel and that they determine the optical properties (Örmo *et al*. 1996; Yang *et al*. 1996). Presently, coordinate data of over 100 GFP-like proteins including mutants have been deposited in the Protein Data Bank. Their chromophore structures can be roughly classified into two groups according to the number of steps involved in chromophore processing. The first includes the avGFP type described and the second includes the DsRed type that undergoes an additional chemical reaction to assume a trans-conformational form between two peptides, enabling the emission of red fluorescence (Gross *et al*. 2000; Yarbrough *et al*. 2001).

Six GFP homologs comprise a new class of avGFP from marine copepods of the family Pontellidae (Shagin

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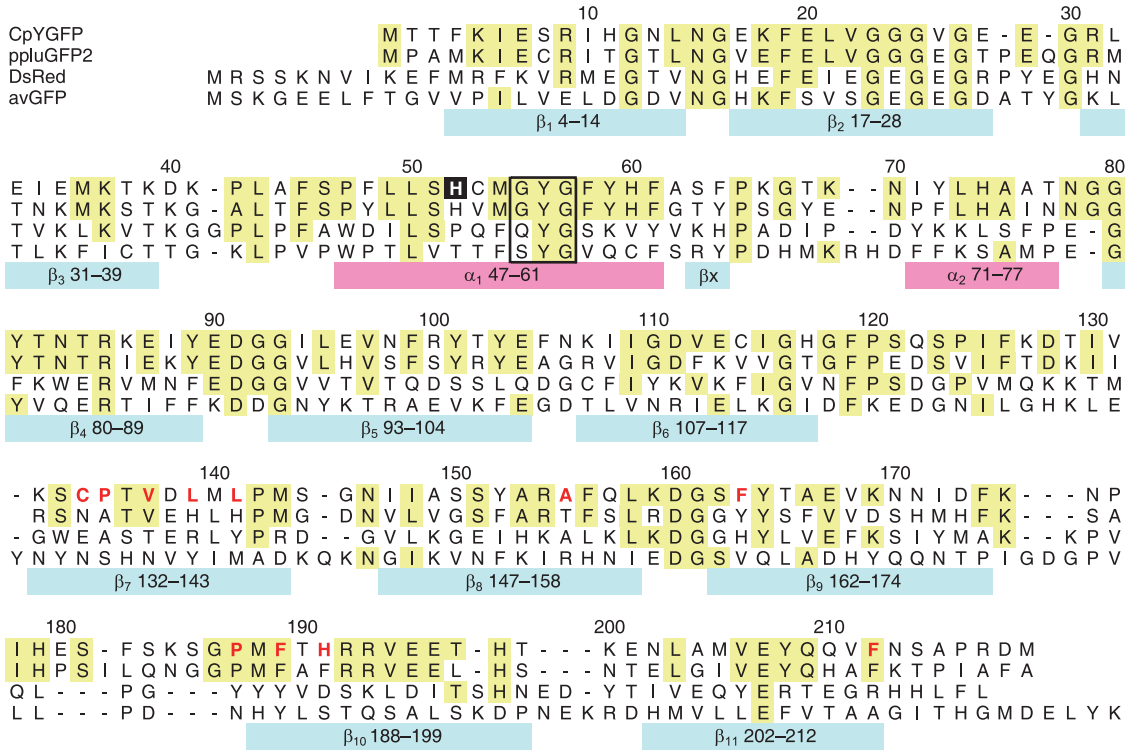


Figure 1 Amino acid sequence alignment of four GFP-like proteins: CpYGFP (Accession No. AB185173), ppluGFP2 (No. AY268072), DsRed (No. AF168419) and avGFP (No. AF183395). Secondary structure of CpYGFP is indicated at bottom of alignment. Regions marked in blue and pink are β-strands and α-helices, respectively. Three boxed residues show chromophore. H in black box is histidine residue involved in stacking interaction with chromophore. Red residues are hydrophobic and used for dimer formation (see also Fig. 2c). Yellow highlight, conserved residues.

et al. 2004) and a homologous protein emitting novel yellowish-green fluorescence has been cloned from the marine copepod *Chiridius poppei* (CpYGFP) (Masuda *et al.* 2006). The expressed protein has excitation and emission maxima at 509 and 517 nm, respectively, which are significantly longer than those of avGFP (395 (475) and 508 nm, respectively) (Morise *et al.* 1974), but much shorter than those of drFP583 (558 and 583 nm) from *Discosoma* sp. (Matz *et al.* 1999), which is commercially available as DsRed. Furthermore, CpYGFP has significantly red-shifted spectra compared with other copepod GFPs despite having high amino acid sequence similarity. The crystal structure of one copepod GFP from the planktonic copepod *Pontellidae plumata* (ppluGFP2) has been determined (Wilmann *et al.* 2006) and its engineered protein commercially available as TurboGFP has also been determined (Evdokimov *et al.* 2006). Here, we used X-ray crystallography to analyze the structure of CpYGFP and then compared it with those of ppluGFP2 and TurboGFP to determine the structural basis for the red-shift in CpYGFP. These investigations revealed a novel stacking interaction between His52 and the phenol moiety of the chromophore. We found that His52 is the fourth residue preceding Tyr56 of the tri-peptide sequence of the chromophore. We also performed mutational studies to confirm whether this stacking interaction is important to determining the optical properties.

Results

Overall structure

While the primary structure of CpYGFP had only 21% identity with DsRed (Fig. 1), the latter functioned well as an initial model during MR calculations and two monomers were identified in an asymmetric unit. Residues 2–218 of one monomer (Monomer A) and residues 2–217 of another (monomer B) were assigned to an electron density map. The first methionine, which could not be found in the map, appeared to be digested because the next residue was threonine. The final model contained one each of monomers A and B, one CAPS

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(*N*-cyclohexyl-3-aminopropanesulfonic acid) and 437 water molecules. The CAPS molecule in the crystal played an important role as an additive for packing insofar as the space group of $C222_1$ would change to $P2_13$ with a much larger unit cell in its absence (data not shown). The CAPS molecule was located between monomers A and B (monomer B and symmetry-related monomer A in PDB code 2DD7) and it interacted with both monomers (Supporting Fig. S1). The cyclohexyl ring of CAPS interacted with a hydrophobic patch formed by Pro124 and Tyr164 and the methylene chain of the Glu90 of monomer B, and the positive charge of the nitrogen atom of CAPS interacted with the Glu90 of monomer B. This nitrogen atom also interacted with the Asp160 of monomer A, and the sulfonic acid moiety in CAPS interacted with Thr129 and Lys127 of monomer A. This CAPS-induced lattice appeared to result in a rather low averaged B-factor (Table 2) despite the high solvent content (70%).

Two A monomers formed a dimer around the crystallographic twofold axis and two B monomers similarly formed another. The root mean square (rms) difference in the Cα superpositions between the two monomers in an asymmetric unit was 0.42 Å for all residues from 2 to 217. Loops flipped in residues from 143 to 146 of one monomer, but not in the other. If the residues from 143 to 146 were not included, then the rms difference decreased to 0.21Å. For simplicity, the following discussion is based on monomer A, unless noted otherwise. The secondary structure of monomer A is β1 (4–14), β2 (17–28), β3 (31–39), α1 including a chromophore region (47–61), βextra (63–64), α2 (71–77), β4 (80–89), β5 (93–104), β6 (107–117), β7 (132–143), β8 (147–158), β9 (162–174), β10 (188–199), and β11 (202–212) (Fig. 1) and folded into a β-barrel motif comprising 11 β-strands and a central helix (Fig. 2a,b), which is a characteristically common fold among GFP-like proteins. The overall folding of CpYGFP is similar to that of other GFP-like proteins, where 11 β-strands enclose a central helix with a chromophore to form a β-barrel structure that is highly conserved (Fig. 2b). For example, the rms difference in the Cα superposition including all strands and helices between CpYGFP and DsRed (Wall *et al.* 2000), avGFP (Örmo *et al.* 1996) or ppluGFP2 (Wilmann *et al.* 2006) is 0.93, 1.24 or 0.94 Å, respectively. To correctly align the amino acid sequences except for ppluGFP2 is difficult before structural determination even using sequence alignment programs because the amino acid sequence identities of CpYGFP to DsRed and avGFP are only 21.0% and 15.5%, respectively, despite the structural similarities. Structure-based sequence alignment (Fig. 1) shows that insertions, deletions and low amino acid identities are apparent in the region from 176 to the C-terminus (CpYGFP numbering) compared with other regions. In particular, a coil-like loop between β9 and β10, which lies near α2 and covers the top of the β-barrel (Fig. 2b), has the most divergent feature among these GFP-like proteins. The conformation at the C-terminal end of this loop in CpYGFP appears to contribute to enlargement of the contact surface for dimerization. On the other side of the β-barrel, the conformation of a loop around residue 200 (CpYGFP numbering) is variable, but this loop of CpYGFP is cut short, and it forms the most compact fold of this region among GFP-like proteins (Fig. 2b).

Dimerization

All known GFP-like proteins form monomers, dimers or tetramers. For example, avGFP is monomeric in aqueous solution, but dimeric at high concentrations and under specific solvent conditions. Crystal structures of avGFP have also been solved as monomers (Örmo *et al.* 1996) and dimers (Yang *et al.* 1996). On the other hand, the red fluorescent DsRed and eqFP611 proteins and nonfluorescent Rtms5 proteins form tetramers, which have a similar 222-symmetry fold through two independent interacting surfaces (Wall *et al.* 2000; Petersen *et al.* 2003; Prescott *et al.* 2003). The present crystal structure of CpYGFP is dimeric, which is in agreement with the value of the molecular mass estimated by gel filtration column chromatography (Masuda *et al.* 2006). Hydrophobic interaction is mainly involved in CpYGFP dimerization. A region consisting of Val137, Phe189, His191 and Phe212 (highlighted with yellow in Fig. 2c) forms the main part of the hydrophobic interface, which interacted with the same region on the neighboring monomer. Another region consisting of Cys134, Pro135, Ala155 and Phe163 interacts with a region including Leu139, Leu141 and Pro187 of the neighboring monomer (highlighted with green in Fig. 2c). Such relative positions of monomers in a dimer of CpYGFP appear different from those of the avGFP dimer (Yang *et al.* 1996), but similar to those of the ppluGFP2 dimer (Wilmann *et al.* 2006) and also to those in a dimer within the DsRed tetramer (Wall *et al.* 2000) as follows. The DsRed tetramer has two independent dimer-interfaces within the 222-symmetry. One is between monomers A and B, and the other is between monomers A and C (Fig. 2d). The former is involved in hydrophilic interactions including water-mediated hydrogen bonds and the latter is involved in hydrophobic interactions mainly due to valine and isoleucine residues. When monomer A of CpYGFP was fitted to monomer A of

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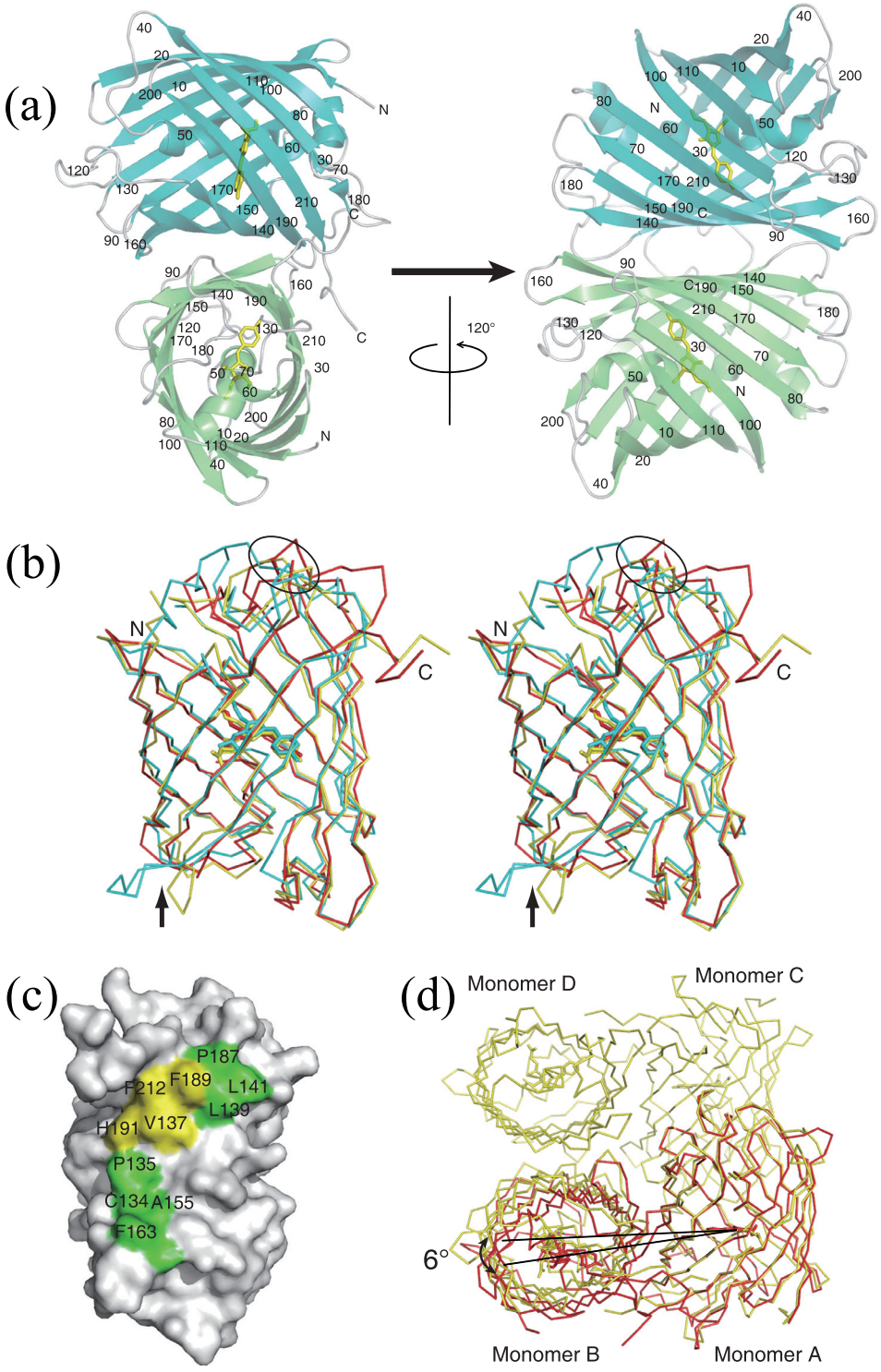


Figure 2 Overall structure of CpYGFP. (a) Ribbon models of dimer and its side view. (b) Stereo view of superposition of CpYGFP structures (red), avGFP (blue) and DsRed (yellow). Loops with large deviations are shown by black circle and arrow. (c) Surface model of inner interface of monomer to form dimer. Hydrophobic residues participating in dimer formation are colored yellow or green. (d) Superposition of CpYGFP dimer onto DsRed dimer within tetramer. When monomer A of CpYGFP is fitted to one tetramer of DsRed, 6° of rotation is required for superposition of monomers B. Colors representing CpYGFP and avGFP are as described in (b).

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DsRed, monomer B of CpYGFP was superposed very closely to monomer B but not to monomer C of DsRed (Fig. 2d). This means that the hydrophobic dimer-interface in CpYGFP is replaced by the hydrophilic dimer-interface in DsRed. This is understandable because most of hydrophobic residues participating in dimerization in CpYGFP are replaced by hydrophilic residues in DsRed as shown in Fig. 1. Therefore, the relative position of the dimer can be considered as highly conserved, whereas the interaction manner of dimerization is not, as far as CpYGFP and DsRed are concerned.

Chromophore structure and environment

The chromophore of CpYGFP consists of a Gly55-Tyr56-Gly57 tri-peptide sequence. Around the chromophore, only Arg85 and Glu207 are conserved, as found in other GFP-like proteins, and they are considered to be involved in the catalysis of chromophore formation, specifically in cyclization and/or oxidation (Matz *et al.* 1999). The side-chain amino group of Arg85 hydrogen bonds to the imidazolinone carbonyl group of the chromophore, while the carboxyl oxygen atom of Glu207 hydrogen bonds to the imidazolinone amino group of the chromophore. The hydroxyl group of the phenol moiety of the chromophore forms a hydrogen bond with the side-chain oxygen atom of Thr136. This hydrogen bond is conserved in ppluGFP2 and TurboGFP. In avGFP, Thr136 is not conserved, but Thr203 might be displaced to maintain the hydrogen bond to the hydroxyl group of the phenol moiety. Water molecules also contribute to stabilization of the chromophore structure, and two of them are involved in direct hydrogen bonding with the chromophore (Fig. 3a). The most notable structural feature around the chromophore is that His52 is stacked on its phenol moiety (Supporting Fig. S2). The shortest atom–atom distance between stacked planes is 3.14 Å in monomer A and 3.18 Å in monomer B. His52 is also stacked on the guanidino group of Arg154, forming a stable sandwiched structure. His52 in CpYGFP is positioned at the fourth residue preceding Tyr56 in the chromophore. If the residues from His52 to Tyr56 were to participate in a typical α-helix, His52 could not stack on the phenol moiety because the pitch of a typical α-helix is known to be 3.6. Hence, the central helix of GFPs have an unusual pitch at the chromophore as a result of an autocatalytic cyclization on the main chain, which then might allow His52 stacking to occur on the chromophore. The residue corresponding to His52 is threonine in some GFP-like proteins including avGFP, or proline in some other GFP-like proteins including DsRed. These residues could not participate in stacking interaction with the phenol moiety of the chromophore. A stacking interaction also occurs in the quadruple mutant T203Y/S65G/V68L/S72A of GFP, referred to as YFP (Wachter *et al.* 1998), where the phenol side-chain of Tyr203 is the stacking residue, although Tyr203 is located at the opposite side relative to the phenol plane of the chromophore in contrast with the case of CpYGFP.

Table 1 Summary of emission and excitation maxima

	Emission max (nm)	Excitation max (nm)
Wild type	517.0	509.0
H52T	511.2	495.4
H52D	511.0	408.2 and 499.2
H52F	521.8	513.6

H52T, H52D, H52F mutants and crystal structure of H52T

Whether or not the wavelength shift is caused by a stacking interaction of His52 was examined by mutation experiments. The reasons why we selected mutant H52T, H52F and H52D proteins were as follows. Threonine in H52T can serve as a representative residue of alanine, valine, leucine, isoleucine, methionine and threonine, which should have no π–π stacking on the phenol moiety of the chromophore. H52F could confirm stacking effects on the phenol moiety. The tyrosine residue was assumed to behave like phenylalanine. We used aspartic acid in H52D to examine the effect of a negative electrostatic charge on the phenol moiety of the chromophore. Glutamic acid has a longer side chain, which could interfere with the negative charge being able to ride on the phenol ring of the chromophore. Table 1 and Fig. 4c shows the fluorescence spectra of the mutants. The emission maximum of H52T being 511 nm indicated a significant blue-shift, whereas that of H52F was at 522 nm, indicating more of a red-shift than that of the wild type. The blue-shift emission might have been due to the absence of a stacking interaction with the chromophore, and red-shift might be attributable to the amount of overlapping π-electrons of phenylalanine being greater than that of histidine. The spectrum contained a shoulder peak for the wild type at 475–485 nm and for H52F at 480–490 nm, but not for H52T and H52D. This indicates a relationship between the shoulder and the stacking interaction. A shoulder at 480–490 nm is also found in the spectrum of YFP, which has a stacking interaction (Wachter *et al.* 1998).

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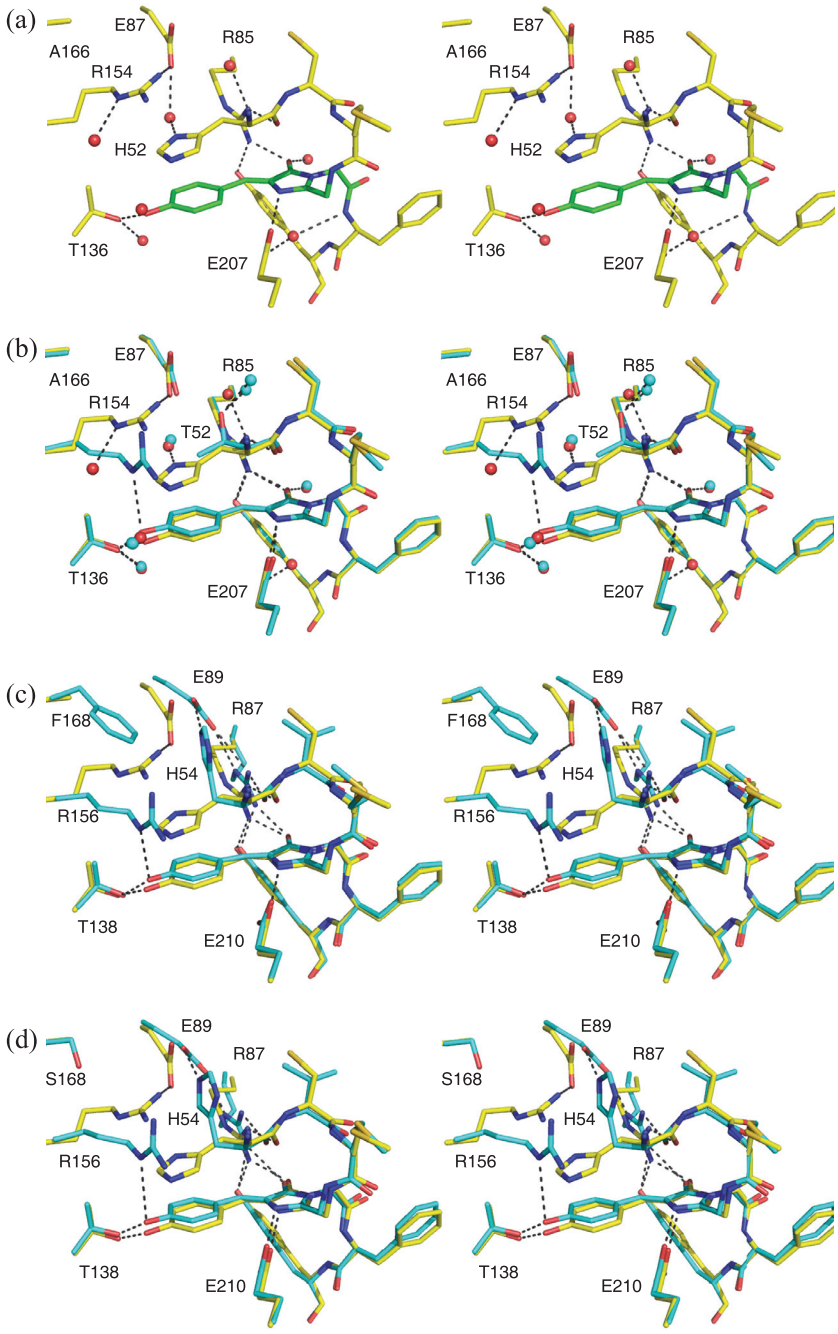


Figure 3 Stereo views of chromophore structure and environment. (a) Green, chromophore; yellow, other residues. Red spheres, water molecules; dashed lines, hydrogen bonds. (b) H52T mutant, (c) ppGFP2 and (d) TurboGFP in blue are superposed on wild-type CpYGFP with yellow. Water molecules in (c) and (d) are omitted for clarity.

We analyzed the crystal structure of H52T to determine how the mutation induces structural changes in the chromophore and its environment. The overall structure was almost identical to that of the wild-type protein insofar as the rms difference of the Cα positions was < 0.2 Å. A single difference is located around the mutated residue near the chromophore (Fig. 3b). Thr52 appears to form CH-π interaction with the phenol moiety of the chromophore in place of π–π interaction. Superposition with the wild type (Fig. 3b) revealed that Arg154 moves close to the chromophore, as if to fill a void created by the replacement of threonine with its smaller side chain. The phenol moiety of the chromophore also moved close to residue Thr52 through a slight tilt (4.5°) of the phenol ring. No other changes were evident. Thus, the structural and mutational studies

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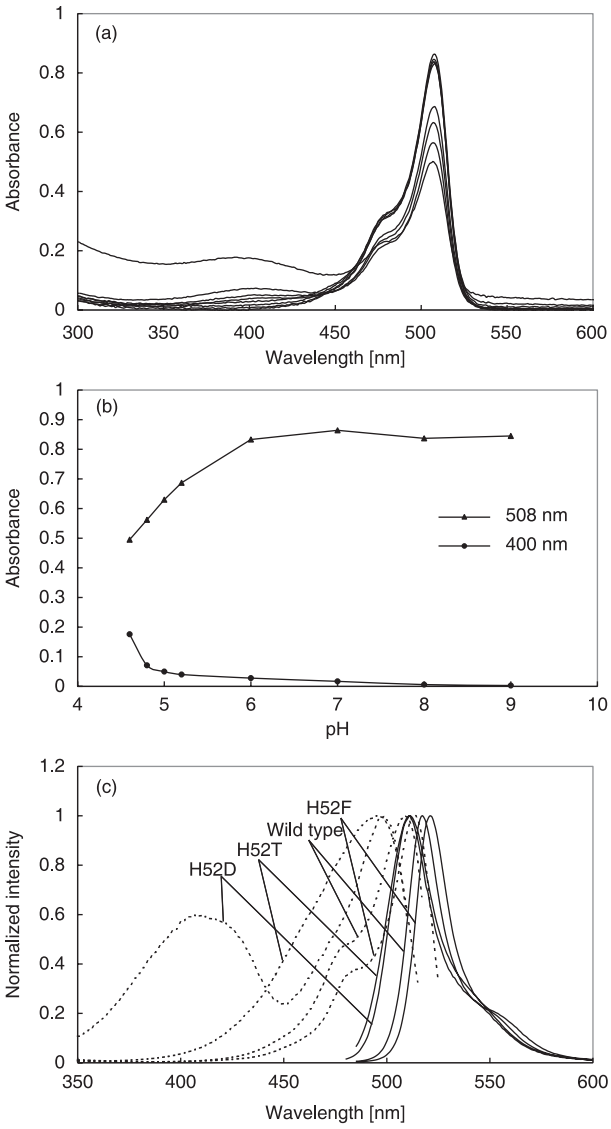


Figure 4 Absorbance and fluorescence spectra of CpYGFP. (a) Absorbance scans of wild-type CpYGFP at various pH values. Eight scans at pH 4.6, 4.8, 5.0, 5.2, 6.0, 7.0, 8.0 and 9.0 are superposed. (b) Absorbance spectra at 508 and at 400 nm are plotted against pH. These spectra were measured at same CpYGFP concentration. (c) Normalized fluorescence excitation (dashed lines) and emission (solid lines) spectra of wild-type CpYGFP, H52T, H52D and H52F.

provided support for the hypothesis that His52 stacking on the phenol moiety of the chromophore induces a red-shift in emission wavelength.

Discussion

Among several fluorescent proteins characterized to date, only two natural proteins other than CpYGFP presented here are known to emit yellow fluorescence (including greenish yellow fluorescent proteins): phiYFP from Hydrozoa with an emission maximum of 537 nm (Shagin *et al.* 2004) and zFP538 from *Zoanthus* with an emission maximum of 538 nm (Matz *et al.* 1999). The crystal structure of zFP538 showed that zFP538 contains a three-ring chromophore derived from a transimination reaction, suggesting that chromophore conjugation by an additional double bond lowers the absorption and emission frequenciess (Remington *et al.* 2005). On the other hand, the crystal structure of phiYFP remains obscure, but phiYFP is a GFP-like protein according to its absorption spectra under conditions of acid denaturation (Shagin *et al.* 2004). Indeed, phiYFP contains tyrosine at position 203, which is the key mutation site indispensable to yellow emission in a commercially available GFP mutant with the substitution T203Y. The crystal structure of YFP (quadruple mutant T203Y/S65G/V68L/S72A of avGFP with an emission maximum of 528 nm) shows that Tyr203 has stacking interaction with the phenol moiety of the chromophore (Wachter *et al.* 1998). In contrast to the stacking interaction in CpYGFP determined herein, the phenol side-chain of Tyr203 is located at the opposite side relative to the phenol plane of the chromophore. Replacing Tyr203 with a histidine leads to emission at 524 nm (Örmo *et al.* 1996), and replacement with a phenylalanine results in emission at 525 nm (Dickson *et al.* 1997). These results are comparable to the present findings generated by replacing His52 with Thr, Asp and Phe (Table 1 and Fig. 4c).

In comparison with previous findings of yellow fluorescent proteins, the phenol moiety of the chromophore of CpYGFP has a novel stacking interaction with His52, which is located four residues before Tyr56 in the chromophore. His52 is not conserved except in two copepod GFPs (ppluGFP2 and TurboGFP), although these proteins emit green fluorescence. This difference in optical properties among copepod GFPs can be explained by the structural comparison shown in Fig. 3c,d. In ppluGFP2, His54 (corresponding to His52 in CpYGFP) is flipped by rotating the Cα-Cβ bond compared with His52 in CpYGFP. This flip disrupts the stacking interaction with the phenol moiety of the chromophore that is essential for red-shifted emission. The bulky side chain of Phe168 in ppluGFP2 (corresponding to residue Ala166 in CpYGFP) is inserted between Glu89 and Arg156, disrupting the salt bridge between the two and causing them to move apart. The flipping of His54 appears to be induced to avoid contact with the rearranged Arg156. The flipped His54 participates in a new hydrogen bond with Glu89 (Fig. 3c). Phe168 with a bulky side

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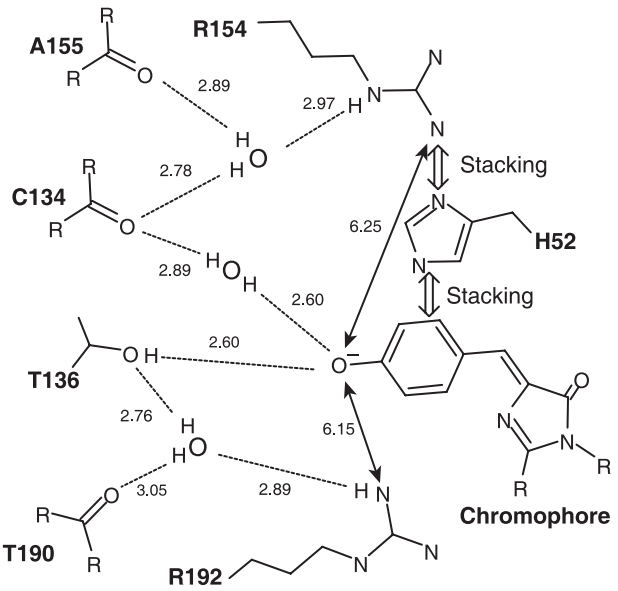


Figure 5 Schematic diagram showing environment of anionic state of the chromophore. Dotted lines: hydrogen bonding networks; double-headed arrows: electrostatic interactions. Positions of hydrogen atoms within hydrogen bonding networks are tentatively assigned.

chain is replaced with Ser in TurboGFP (Evdokimov *et al*. 2006). However, the structural rearrangement including His54 flipping is similar even with this replacement (Fig. 3d). Such structural rearrangement caused by the amino acid displacement might be due to the tight packing around Ala166 in CpYGFP.

Figure 4c and Table 1 show that the wild type has a single lower energy peak as well as H52T or H52F, whereas H52D has two excitation maxima at 408 and 499 nm. Two similar absorption maxima have been identified for avGFP at 395 and 475 nm (Morise *et al*. 1974); the former and latter wavelengths may be ascribed to the neutral and anionic states of the chromophore respectively but the mutation of avGFP (S65T) completely suppresses the 395 nm peak. A structural comparison of avGFP and the S65T mutant suggested that the equilibrium between the anionic and neutral states is governed by a hydrogen bonding network that permits proton transfer between the chromophore and neighboring side chains (Brejc *et al*. 1997; Shu *et al*. 2007). The present findings indicated that the chromophores of the wild type, H52T and H52F should be anionic. Hydrogen bonding networks (Fig. 5) suggest that the hydroxyl group of the phenol moiety is deprotonated and that it acts as a proton acceptor from Thr136. Each of Arg154 and Arg192 participates in hydrogen bonding networks and also functions as a positively charged residue that is important for electrostatic interaction with the anionic phenol moiety. The positively charged environment formed mainly by Arg154 and Arg192 contributes to stabilization of the anionic form of the chromophore. The above networks and arginine residues are conserved in ppluGFP2 and TurboGFP. This perspective is consistent with that the estimated pKa is ≤ 4.6 based on apparent titration curves generated from absorbance scans (Fig. 4b) and the reported pKa for ppluGFP2 is 4.3, which is the lowest value for a GFP that can be excited at ~488 nm (Wilmann *et al*. 2006). Figure 4b shows that the wild type of CpYGFP has one absorbance peak at 508 nm when the pH is > ~4.8 in the titration curve, which reflects that the chromophore is in a highly stabilized anionic state. This is in contrast to the spectral properties of YFP that are highly sensitive to pH change (Wachter *et al*. 1998). On the other hand, H52D has two excitation peaks (Table 1), suggesting the presence of two different forms caused by the neutral and anionic states of the chromophore, as in the case of avGFP mentioned above. The protonated form of the phenol moiety of the neutral chromophore might be occurred even in the wild type under acidic condition because of observation of small peaks around 400 nm under pH4.6 and pH4.8 (Fig. 4a).

The present study and comparison of GFP-like proteins from a planktonic copepod showed that His52 is a key residue in the determination of emission wavelength because stacking interaction of this residue with the chromophore is likely to direct a red-shift. Spectral properties are changed when His52 is replaced. Specific properties engineered within copepod GFP-like proteins might be valuable for use in biotechnology such as by functioning as reporters in physiological studies. Moreover, even other GFP-like proteins, mutation to Tyr, Phe or His at the position corresponding to His52 in CpYGFP, that is, at the fourth residue preceding Tyr within the chromophore could lead to stacking interaction with the chromophore. Because this position is easily identified compared with another position corresponding to Y203 in YFP far from the chromophore on sequence, engineering at the former position could be performed effectively for spectroscopic alterations.

Experimental procedures

Protein preparation and characterization of mutants

Recombinant CpYGFP was expressed and purified as described (Masuda *et al*. 2006). Plasmids overexpressing CpYGFP mutants (H52T, H52F and H52D) were prepared by site-directed mutagenesis and expressed in *Escherichia coli* BL21-CodonPlus

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Table 2 Data collection and refinement statistics for CpYGFP and CpYGFP H52T

	Wild type	H52T
Data collection statistics		
Wavelength (Å)	0.9799	1.0000
Resolution range (Å)	30–1.9 (2.00–1.90)	50–2.3 (2.42–2.30)
Number of measurements	383 411 (55 090)	538 173 (79 203)
Number of reflections	65 109 (9438)	74 303 (10 756)
Completeness (%)	99.9 (100.0)	100.0 (100.0)
R_{merge}†	0.073 (0.291)	0.103 (0.354)
$I/\sigma(I)$	18.2 (5.2)	18.1 (6.7)
Refinement statistics		
Resolution range used (Å)	30–1.90	30–2.3
Number of reflections working (free)	65 081 (3281)	70 447 (3759)
R_{work}‡	0.204	0.184
R_{free}‡	0.231	0.217
Number of protein atoms/AU	3442	6880
Number of solvent atoms/AU	451	560
rms deviation in bond length (Å)	0.0081	0.0083
rms deviation in bond angle (°)	1.53	1.50
Average B-factor (Å^2)	21.2	20.9
B-factor on Wilson plot (Å^2)	12.7	23.6

Values in parentheses refer to the highest resolution shell.
†$R_{merge} = \Sigma_h\Sigma_i \left| I_{h,i} - \langle I_h \rangle \right| / \Sigma_h\Sigma_i \left| I_{h,i} \right|$, where h refers to unique reflection indices and i indicates symmetry-equivalent indices.
‡R values calculated as $\Sigma| |F_o| - |F_c| | / \Sigma |F_o|$, where F_o and F_c are the observed and calculated structure factor amplitudes.

(DE3)-RIL Competent Cells (Stratagene). Protein samples were purified by anion exchange column chromatography, using HiTrap DEAE (GE Healthcare), and the absorbance of the eluants was monitored at 490 nm. The proteins were concentrated using VIVAspin (Sartorius) and diluted with 20 mM Tris-HCl (pH 8.0) containing 0.2 M NaCl. Emission spectra were measured using an F-4500 fluorescence spectrometer (Hitachi) at 480 or 470 nm, whereas excitation spectra were measured at 520 (wild type and H52D), 518 (H52T) and 530 (H52F) nm. The fluorescence spectra in Fig. 4c are normalized from their peaks, and Table 1 summarizes the peak wavelengths for each protein. Protein solutions used for crystallization were further purified using Superdex 200 10/30 GL gel filtration and by anion exchange chromatography, using MonoQ (GE Healthcare).

Crystallization and structure determination

Purified CpYGFP was crystallized by sitting-drop vapor diffusion at 18 °C. Drops comprised 1 μL of 20 mM Tris-HCl (pH 8.5) containing 80 mM NaCl and 23 mg/mL of CpYGFP with 1 μL of reservoir solution. Rod-shaped crystals formed in reservoir solution comprising 0.1 M *N*-cyclohexyl-3-aminopropanesulfonic acid (CAPS) (pH 10.5), 0.2 M Li_2SO_4 and 2.2 M $(NH_4)_2SO_4$. Diffraction data were collected under cryogenic conditions at 100 K under a stream of nitrogen at beamline BL-6A of the Photon Factory (PF; Tsukuba, Japan) using an ADSC Quantum-210 CCD detector. The cryoprotectant comprised a mixture of glycerol and reservoir solution. The space group was $C222_1$ with a unit cell of a = 113.5, b = 133.5 and c = 108.7 Å, which corresponded to V_M = 4.2 Å^3/Da (solvent content 70%) when the asymmetric unit contained two monomers. The reflection data to 1.9Å resolution was processed using the MOSFLM program (Leslie 2006). We used 20 mM Tris-HCl (pH 8.5), containing 100 mM NaCl and 8 mg protein/mL to crystallize H52T, and the optimal reservoir solution, 0.1 M CAPS (pH 10.5), containing 0.2 M NaCl and 2.6 M $(NH_4)_2SO_4$. Data sets were collected using an ADSC quantum 210 CCD detector at beamline NW12, PF-AR (Tsukuba, Japan). The space group was $P2_12_12_1$ with the unit cell being a = 108.7, b = 114.0 and c = 133.4 Å. The unit cell constants were similar to those of wild-type crystals except the crystal had another space group. The V_M value was 4.2 Å^3/Da when the asymmetric unit contained four monomers. Table 2 summarizes the data collection statistics.

Initial phases of CpYGFP were calculated using molecular replacement (MR). A search model was constructed based on coordinates of DsRed deposited in the Protein Data Bank with an ID of 1GGX (Wall *et al.* 2000). We performed structural refinements using the CNX (Brunger *et al.* 1998) and XtalView programs (McRee 1999). We omitted 5% of the reflection data when calculating the free R-factor during refinement. The final R_{work} was 20.4% (R_{free} = 23.1%). A phase set for the mutant crystal was calculated using MR and the structure of wild-type CpYGFP to refine the model. Table 2 summarizes the refinement statistics. The atomic coordinates and structure factors for wild-type

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CpYGFP (code 2DD7) and the H52T mutant (code 2DD9) have been deposited in the Protein Data Bank, Research Collaboratory for Structural Bioinformatics, Rutgers University, New Brunswick, NJ (http://www.rcsb.org).

Measurement of absorbance spectra

Concentrated CpYGFP in 5 mM HEPES-NaOH (pH 7.3) and 0.1 M NaCl was diluted into measurement buffers comprising 0.1 M acetate-NaOH (pH 4.6, pH 4.8, pH 5.0, pH 5.2), MES-NaOH (pH 6.0), Tris-HCl (pH 7.0, pH 8.0) or Bicine-NaOH (pH 9.0) and 0.1 M sodium chloride. CpYGFP and measurement buffer mixed at a ratio of 14:3 were incubated at room temperature for 40 min and then absorbance spectra were measured using an Ultrospec 2100 pro (GE Healthcare).

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Supporting Information/Supplementary material

The following Supporting Information can be found in the online version of the article:

Figure S1 *N*-Cyclohexyl-3-aminopropanesulfonic acid (CAPS) located between monomers B and A' (symmetry-related monomer A) of CpYGFP. (A) Ribbon model. (B) Schematic presentation of CAPS involved in hydrophilic and hydrophobic interactions with residues in monomers B and A'.

Figure S2 Stereo view of $2F_o$-F_c electron density map contoured at 2.0 σ cutoff at the chromophore with His52 and Arg154 in monomer A.

Additional Supporting Information may be found in the online version of this article.

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